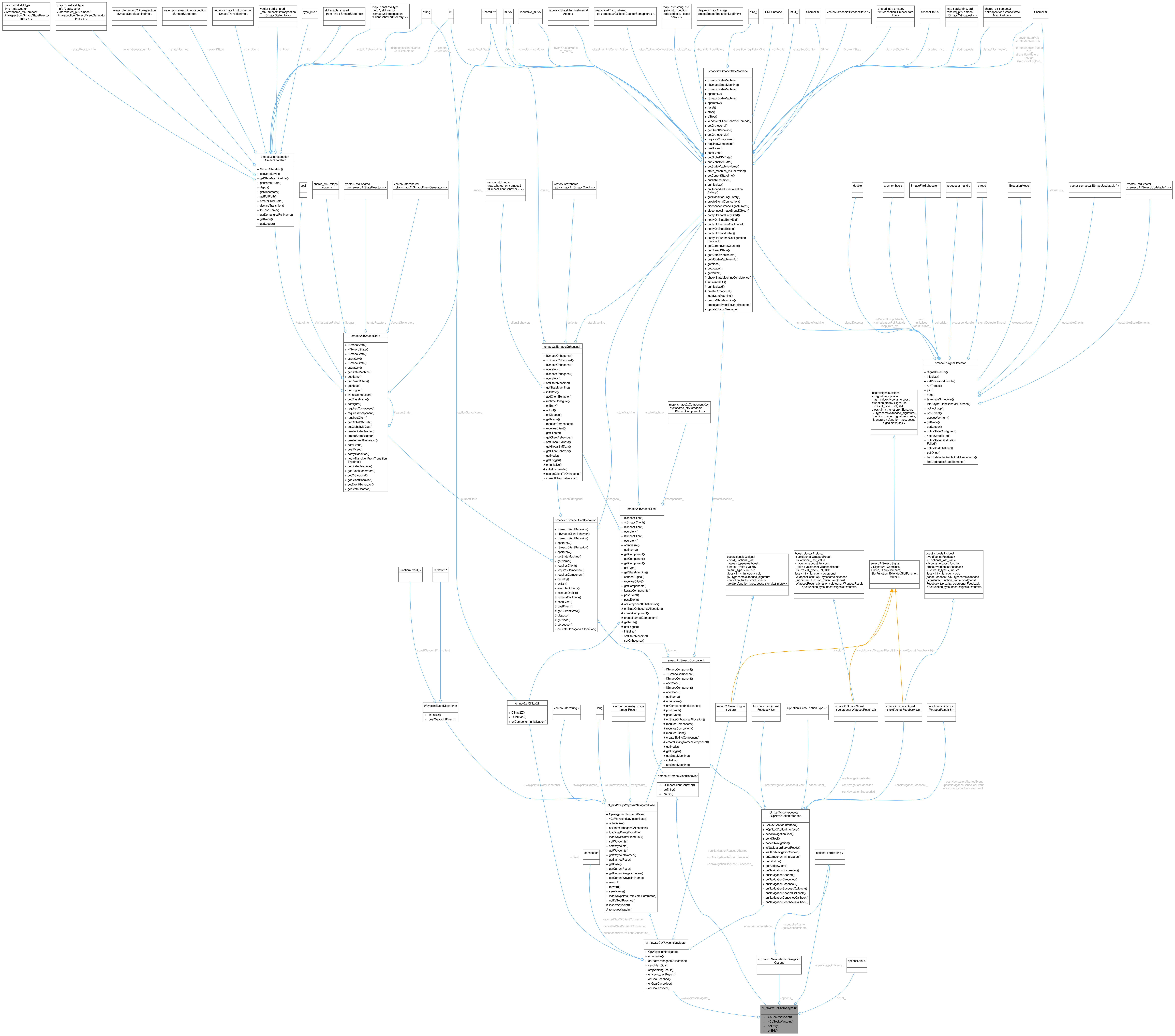
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
weak_ptr< smacc2::introspection ::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection ::SmaccStateInfo >
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
string
int
SharedPtr
mutex
recursive_mutex
atomic< StateMachineInternal Action >
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
deque< smacc2_msgs ::msg::SmaccTransitionLogEntry >
size_t
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2 ::introspection::SmaccState Info >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
SharedPtr
+stateReactorsInfo
+eventGeneratorsInfo
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+staticBehaviorInfo
+demangledStateName
+fullStateName
+depth_
+stateIndex_
-reactorWalkDepth_
#nh_
-transitionLogMutex_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-transitionLogHistorySize_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
#eventsLogPub_
#stateMachinePub
_
#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ ISmaccStateMachine()
+ operator=()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ onUnhandledStInitialization Failure()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
bool
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
#node_
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
-mutex_
vector< std::shared _ptr< smacc2::ISmaccClient > >
double
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
-statusPub_
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
#stateInfo_
#initializationFailed_
#logger_
#stateReactors_
#eventGenerators_
-clientBehaviors_
#clients_
-stateMachine_
-smaccStateMachine_
-signalDetector_
-kDefaultLoopRateHz
-kInitializationPollRateHz
-loop_rate_hz
-end_
-initialized_
-rosInitialized_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
smacc2::ISmaccState
+ ISmaccState()
+ ~ISmaccState()
+ ISmaccState()
+ operator=()
+ ISmaccState()
+ operator=()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ initializationFailed()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::ISmaccOrthogonal
+ ISmaccOrthogonal()
+ ~ISmaccOrthogonal()
+ ISmaccOrthogonal()
+ operator=()
+ ISmaccOrthogonal()
+ operator=()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyStateInitialization Failed()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
-actionServerName_
-stateMachine_
-stateMachine_
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
-currentState
-currentOrthogonal
-orthogonal_
#components_
#stateMachine_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ ISmaccClientBehavior()
+ operator=()
+ ISmaccClientBehavior()
+ operator=()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ ISmaccClient()
+ operator=()
+ ISmaccClient()
+ operator=()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const WrappedResult &), optional_last_value < typename boost::function _traits< void(const WrappedResult &)>::result_type >, int, std:: less< int >, function< void(const WrappedResult &)>, typename extended _signature< function_traits< void(const WrappedResult &)>::arity, void(const WrappedResult &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(const Feedback &), optional_last_value < typename boost::function _traits< void(const Feedback &)>::result_type >, int, std ::less< int >, function< void (const Feedback &)>, typename extended _signature< function_traits< void(const Feedback &)>::arity, void(const Feedback &)>::function_type, boost::signals2::mutex >
function< void()>
ClNav2Z *
#owner_
< void()>
< void(const WrappedResult &)>
< void(const Feedback &)>
+postWaypointFn
+client_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ ISmaccComponent()
+ operator=()
+ ISmaccComponent()
+ operator=()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
WaypointEventDispatcher
+ initialize()
+ postWaypointEvent()
cl_nav2z::ClNav2Z
+ ClNav2Z()
+ ~ClNav2Z()
+ onComponentInitialization()
vector< std::string >
long
vector< geometry_msgs ::msg::Pose >
smacc2::SmaccSignal < void()>
function< void(const Feedback &)>
CpActionClient< ActionType > *
smacc2::SmaccSignal < void(const WrappedResult &)>
smacc2::SmaccSignal < void(const Feedback &)>
function< void(const WrappedResult &)>
smacc2::SmaccClientBehavior
+ ~SmaccClientBehavior()
+ onEntry()
+ onExit()
+waypointsEventDispatcher
#waypointsNames_
+currentWaypoint_
#waypoints_
+postNavigationFeedbackEvent
-actionClient_
+onNavigationAborted
_
+onNavigationCancelled
_
+onNavigationSucceeded_
+onNavigationFeedback_
+postNavigationAbortedEvent
+postNavigationCancelledEvent
+postNavigationSuccessEvent
cl_nav2z::CpWaypointNavigatorBase
+ CpWaypointNavigatorBase()
+ ~CpWaypointNavigatorBase()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ loadWayPointsFromFile()
+ loadWayPointsFromFile2()
+ setWaypoints()
+ setWaypoints()
+ getWaypoints()
+ getWaypointNames()
+ getNamedPose()
+ getPose()
+ getCurrentPose()
+ getCurrentWaypointIndex()
+ getCurrentWaypointName()
+ rewind()
+ forward()
+ seekName()
+ loadWaypointsFromYamlParameter()
+ notifyGoalReached()
# insertWaypoint()
# removeWaypoint()
cl_nav2z::components ::CpNav2ActionInterface
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()
connection
+client_
+onNavigationRequestAborted
_
+onNavigationRequestCancelled
_
+onNavigationRequestSucceeded_
optional< std::string >
-abortedNav2ZClientConnection
_
-cancelledNav2ZClientConnection
_
-succeededNav2ZClientConnection_
+nav2ActionInterface_
+controllerName_
+goalCheckerName_
cl_nav2z::CpWaypointNavigator
+ CpWaypointNavigator()
+ onInitialize()
+ onStateOrthogonalAllocation()
+ sendNextGoal()
+ stopWaitingResult()
- onNavigationResult()
- onGoalReached()
- onGoalCancelled()
- onGoalAborted()
cl_nav2z::NavigateNextWaypoint Options
-seekWaypointName_
optional< int >
+waypointsNavigator_
+options_
-count_
cl_nav2z::CbSeekWaypoint
+ CbSeekWaypoint()
+ ~CbSeekWaypoint()
+ onEntry()
+ onExit()